Where Shortness and Earliness Compete:
Processing of Fronted Complex-Wh-NPs in Japanese

Under active dependency formation hypothesis [1, 2], "locality" can be divided into two senses: structural closeness (*shortness*) and linear distance (*earliness*) [3, 4, 5].

(1) [[… Filler…] [… Gap-1 …] … Gap-2 …]]

In (1), Gap-1 is linearly close to the filler, but Gap-2 is structurally close to it. A crucial question is to what extent the parser prefers one of the two dependencies. The present study this issue using L-maze task [6, 7] to test which dependency the parser predominantly forms in the case where they compete. We focus on a strictly head-final language Japanese, where the distance between a filler and gap is generally ambiguous until the parser hits the later parts of a sentence [8, 9, 10].

Along this line, the previous study [8] focused on a configuration in (2), where an *wh*-NP is fronted into the sentence initial position and licensed by a question particle in its original position.

(2) [[Wh-NP]i $[_{\text{Matrix Clause}}$ NP_{Subj} $[_{\text{Embedded Clause}}$ NP_{Subj} t_i V-Q] V]]

They demonstrated that the wh-NP is firstly attached to the matrix clause (shortest position), and then reattached to the embedded clause (earliest position) by means of *Typing Mismatch Effects* (TMEs) [8, 9] and Filled Gap Effects [1, 2]. However, it is still unclear which property *wh*-dative NPs have requires which effects or more specifically which dependency. This is because the wh-filler is either a *wh*-filler and a thematic filler. Thus, the study identifies the distinction using fronted Complex NPs, where *wh*-element and fronted NP are not fully overlapped in terms of constituency ("Wh-NP_{obj}" and "NP_{Dat}").

(3) [$[_{\text{Complex-NP}}$ [NP_{Subj} [**Wh-NP_{Obj}** V]] **NP_{Dat}**]$_i$ $[_{\text{Matrix Clause}}$ NP_{Subj} $[_{\text{Embedded Clause}}$ NP_{Subj} t_i V-Q] V]]

24 items sets were created. A 2 × 2 × 2 design manipulated wh-element in a fronted Complex NP ("aru" vs. "dono"), gap in an embedded clause, and a particle in embedded clauses ("ka" vs. "to"). For *Wh*-conditions, if the *wh*-filler requires earliest dependency formation, it is expected that TMEs are observed in the non-Q condition. For *No-gap* (filled gap) conditions, if the fronted NP requires firstly shortest dependency, and then reanalysis converts it to the earliest position (i.e., embedded clause), Filled Gap Effects should occur in the embedded clause.

We collected reading time data from 75 Japanese native speakers. The linear mixed effects model analyzing revealed that <u>TMEs were statistically significant regardless of whether the gap in the embedded clause is filled</u> and <u>no Filled Gap Effects were observed in the embedded clause</u>. At the embedded verb region (region 8), the reading times were significantly longer in the Q-conditions ($p < .003$). But, at the final region (region 9), where the matrix verbs appear, the RTs were significantly longer only in the Q-conditions ($p < .02$). At the pre-verbal position (region 7), there were no significant effects in any conditions ($p = .38$).

They indicate that (a) wh-fillers predominantly prefer earliest dependency formation, and (b) fronted dative NPs were not reconstructed into the embedded clause. As for (b), the obtained results are compatible with the idea that dative NPs are firstly posited in the matrix clause [10]. Thus, we conclude that reanalysis operation in [8]'s sense does not occur by default, and that might be specific to the case of wh-dative NP as in (2). These results further suggest that the [8]'s results can be reinterpreted as follows: when shortest dependency and earliest dependency compete in a single constituent the parser prefers the latter to the former. If this is on the right track, then we can argue that there some cases where structure-based parsing is overridden by processing requirement that grammatical dependency should be created "as soon as possible" rather than "as short as possible".

Materials.

Fronted Wh-Complex-NPs (marked with the dative maker *-ni*)

(A). Wh/ Non ("dono" vs. "aru") + Particle type in an embedded clause (Q "ka" vs. Declarative "to")

[$_{\text{Filler-1}}$ Gakusei-ga [$_{\text{Filler-2}}$ **dono/aru** hon]-o okutta kyoju]-ni hisyo-wa Shota-no tomodachi-ga

student-NOM wh-/det book-ACC sent professor-DAT secretary-TOP Shota-GEN friend-NOM

toshokan-de atta ka/to tazunemashita ka?

library-LOC met Q/Declarative asked Q

(B). Wh/ Non ("dono" vs. "aru") + Particle type in an embedded clause (Q "ka" vs. Declarative "to")

[$_{\text{Filler-1}}$ Gakusei-ga [$_{\text{Filler-2}}$ **dono/aru** hon]-o okutta kyoju]-ni hisyo-wa tomodachi-ga

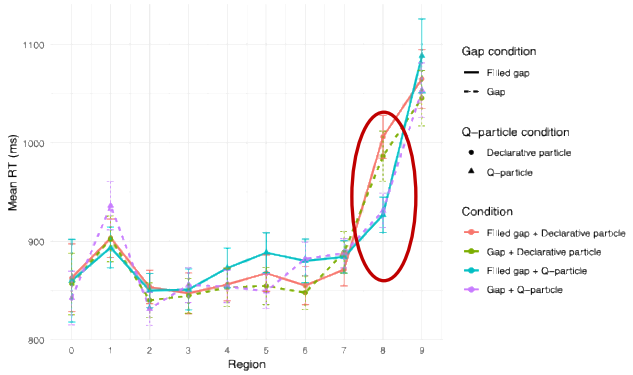
student-NOM wh-/det book-ACC sent professor-DAT secretary-TOP friend-NOM

toshokan-de <u>sisyo-ni</u> atta ka/to tazunemashita ka?

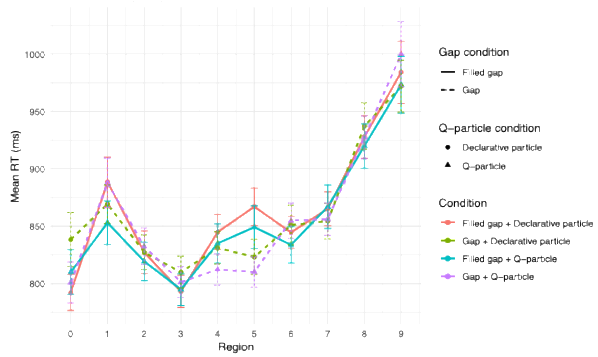
library-LOC librarian-DAT met Q/Declarative asked Q

Figures.

Wh-conditions



Non-wh-conditions



Figures 1 and 2. The time-course reading times in the Wh (left) and the non-Wh (right).

References:

[1] Crain & Fodor (1985), *NELS* 15. [2] Stowe (1986), *Language and Cognitive Processes, 1*(3). [3] Gibson (1998), *Cognition* 68. [4] Warren & Gibson (2002), *Cognition* 85. [5] Chacon et al (2016), *Fronters in Psychology* 7. [6] Forster et al (2009), *Behavior Research Methods*, *41*(1). [7] Witzel & Witzel (2016), *Journal of Psycholinguistic Research, 45*(3). [8] Aoshima et al (2004), *Journal of Memory and Language*, *51*. [9] Takashi & Miyamoto (2003), Ms. [10] Kamide & Mitchell (1999), Language and Cognitive Processes, *14*(5-6).